NAVEENRAJ S.S

Contact: +91 82489 35147
Gmail: rajn51174@gmail.com
LinkedIn: Naveenraj SS
GitHub: NAVEENKCG
Portfolio: https://naveen007-portfolio.vercel.app
Location: Thiruchendur, Tamil Nadu

ELECTRONICS & COMMUNICATION ENGINEER · EMBEDDED SYSTEMS & PCB DESIGNER

CORE SKILLS

- Embedded C / C++
- ESP32 & Arduino
- IoT System Design
- PCB Design
- Embedded Automation
- Sensor Integration
- Python & AI/ML (TinyML)
- UI / Web Design

TECH STACK

Microcontrollers
ESP32 · ARM Cortex-M · Arduino

Hardware & Analog
Op-Amps · Rectifier & Filter Design . RF Modules · FPGA

PCB Design
KiCad · Altium

Protocols & Wireless
UART · BLE · Wi-Fi
ESP-NOW · SPI · I²C

AI / Vision
YOLOv5 · OpenCV
TensorFlow Lite · PyTorch
NumPy · Scikit-learn

Languages
Embedded C · C++ · Python

UI / Web Design
React · Tailwind CSS
Framer Motion · HTML5 · CSS3

CERTIFICATIONS

FPGA Architecture
GUVI / HCL

IoT Boot Camp

PROFILE

3rd-year ECE student with hands-on experience in embedded systems and PCB design. Designed PCBs from schematic to Gerber output using KiCad, Altium, and developed IoT and AI-based embedded projects using ESP32, Embedded C, and TensorFlow Lite. Passionate about R&D in embedded systems and looking for an internship to apply these skills in a real-world engineering environment.

EDUCATION

B.E. — Electronics & Communication Engineering
KCG College of Technology, Chennai · Sep 2024 – Present

Sem 4 · CGPA: 8.15 / 10.00

PROFESSIONAL EXPERIENCE

Industrial Trainee — Data Networking
Bharat Sanchar Nigam Limited (BSNL), Chennai · 1 Week · 2024

- Completed structured training on routing protocols, switching architectures, and enterprise communication systems.
- Gained firsthand exposure to large-scale telecom infrastructure and industrial networking setups.

KEY PROJECTS

RF Defence System — PCB Design & Embedded | *Jan – Mar 2026*

Stack: KiCad · ESP32 · LM358 Op-Amp · Embedded C

- Designed a full RF interference detector PCB from schematic to Gerber/DRL outputs using KiCad; integrated LM358 Op-Amp with ESP32 to detect and directionally track anomalous wireless signals.

GitHub: github.com/NAVEENKCG/RF_Defence_System_PCB

Smart Traffic Control — IoT & Embedded Automation | *Jul – Sep 2025*

Stack: Python · YOLOv5 · OpenCV · ESP32

- AI-driven embedded system using YOLOv5 for real-time vehicle detection; ESP32 GPIO controls relay-switched signal timings dynamically based on live traffic density.

GitHub: github.com/NAVEENKCG/Smart_Traffic_ctrl

Posture Detection & Thrombosis Prevention — IoT Sensor System | *Nov 2025 – Feb 2026*

Stack: Sensors · Microcontroller · Embedded C · Python

- Real-time posture monitoring system using multiple sensors and a microcontroller; sensor data is classified on-device and alerts are pushed wirelessly to the user within < 100 ms.

GitHub: github.com/NAVEENKCG/Posture_Detection

KCG Centre of Excellence

Data Networking
BSNL, Chennai

Arduino Programming
GUVI / HCL

ACHIEVEMENTS

Coordinator — Young Technocrats 3.0
Led KCG-wide innovation summit

Hackathonix '26
Built BookBridge (full-stack) in 48 hrs

SCIFEST '25 — Ideathon
KCG Innovation Council, National Science Day

IoT Boot Camp
Accident Rescue System — KCG ECE CoE

Orbit AI — Mind-Controlled Assistive System (BCI) | *In Development*

Stack: ESP32 · ADS1299 · TensorFlow Lite · BLE · Wi-Fi · ESP-NOW

- Wearable non-invasive BCI: EEG signals acquired via ADS1299, classified on-device using TinyML on ESP32/ARM Cortex-M, and delivered over BLE/Wi-Fi/ESP-NOW with < 50 ms latency to control wheelchairs and smart home devices.

GitHub: github.com/NAVEENKCG/Mind_Wave_AImodel

WEB DESIGN

Designed and built responsive UI layouts using React, Tailwind CSS, and Framer Motion animations:

- MindWave Web — UI dashboard for BCI system data visualisation.
- BookBridge — Full-stack book-sharing platform designed and built in 48 hours at Hackathonix '26.
- Portfolio Sites — Personal and client portfolios with animated, responsive layouts.

GitHub: github.com/NAVEENKCG